This Step or From Here On? Crossed Same-State Interventions for Evaluating Quantized Agents

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Abstract

When a quantized agent and its BF16 reference reach similar average task success, but the quantized agent takes a different action at some state, should BF16 take over, and if so for this one step or from here on? Our results indicate that the answer depends on the state and on the controller that continues, so average success, action divergence and the gain from precision replacement need to be measured separately. We propose a crossed same-state intervention: at a frozen, bit-exactly replayable state, we cross the source of the current action (BF16 or quantized) with the controller of the remaining episode. This yields three replacement gains on the same states: the current action only (G), the continuation only (H), or both (F). We study Qwen3-4B and Qwen3-8B under simulated weight and KV-cache quantization with BF16 compute, one greedy user simulator, and forks at the first decision where the two models' actions differ in tool structure (text-only differences are skipped). In τ^2-bench (85 tasks, 20 nonce blocks), replacing a W4 agent's action there changes the task outcome in 15.5–20.9% of task-block units, in both directions, while the net gain is between -1.9 and $+2.9$ pp. In about a quarter of the divergent states that each W4 cell samples (pooled 26.9%; new-task interval 22.1–32.0%), the outcome effect of the same action swap changes with the controller that continues. In this deterministic setting, this share includes sensitivity of the continuation to any change of controller; no same-precision perturbation control separates the two. Strict sign reversals are only 1.2–4.2% of states in the W4 and KV8 cells, and, in the pooled development W4 states, dependence in the two directions is about equally frequent, so the mean interaction is near zero there (not in the post hoc telecom holdout analysis). The three replacement types therefore need to be compared pairwise on the same states; our results do not establish an ordering among them that transfers across configurations, tasks and user simulators. At outcome-independent states, most outcome-changing differences under our user simulator are text differences with the same tool structure. These outcome labels did not reproduce reliably under an alternative, stochastic relay simulator. Quantized-model uncertainty at the start of an action is associated with whether the two models' actions differ (AUC 0.66–0.91, development block); a frozen uncertainty trigger captured more outcome-changing decisions than matched random selection at two of three rates on later nonce blocks of the same tasks, but has not been shown to select net-beneficial replacements. On 34 unseen telecom tasks, replacing the first tool-structure-divergent action of three W4 cells (weighted $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$) has an estimated effect of -4.6 pp (pre-specified crossed 90% CI -7.2 to -2.0); the frozen rule reads the primary endpoint as inconclusive (neither the positive-gain nor the small-effect reading is met), while both pre-registered secondaries (two-sided consequences, fragility repeatability) pass.

1 Introduction

Quantized LLM agents are usually evaluated by average task success next to the full-precision reference, and in closed-loop benchmarks such as τ^2-bench (Barres et al., 2025) that comparison can look reassuring. On our checkpoints, quantized-minus-BF16 success ranges from -0.2 to $+4.2$ pp across four W4 cells. Its new-task (crossed) intervals are about $\pm$6–7 pp wide, so an equivalence test at ±3 pp fails in every cell (Appendix B). Yet the quantized agent (Q) still takes actions that its BF16 reference (B) would not. If B is available as a fallback, as in dynamic precision routing (Li et al., 2026), a practical question arises at every such state: **is it worth letting BF16 take over, for just this step or from here on?**

Judging the two actions does not answer this. Consider a state from our data (q4 AWQ-W4, τ^2-bench airline task 31, nonce block f13, decision 1 of 12). Q looks up a reservation id that does not exist and receives "not found"; B looks up the user's profile. B's action plainly looks better. We froze this state and ran all four combinations of {B's or Q's action} × {B or Q continuing}. Only one succeeds: B's action *followed by* B. If B supplies only this step and Q continues, Q searches for flights, writes a change and fails. If Q's failed lookup stands and B continues, B also fails. Here, the better-looking action has value only under one continuation. **This case illustrates why average success, action divergence and precision-replacement gain should be measured separately; R1 and R2 quantify how often it matters.**

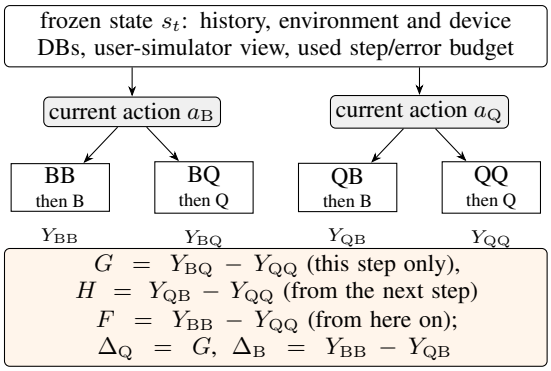


Figure 1: Crossed same-state intervention (schematic, not data). B = BF16 agent, Q = quantized agent. In branch XY, the first letter is the source of the current action and the second is the controller from the next step to the end. Every branch starts from the same frozen state with the original remaining budget. QQ re-executes Q's original episode, and its bit-exact reproduction is the no-op check. Definitions are in §3.2.

What we do. At a frozen state whose no-op replay is bit-exact, we cross the source of the current action with the controller of the remaining episode (Figure 1). Three paired gains answer the three escalation questions on the same states: G replaces only the current action, H only the continuation, F both (§3). We apply the design to two state banks: first tool-structure divergences between B and Q along Q's trajectory, and ordinary states drawn independently of any outcome.

What we find.

- **R1, replacing the current action** (§4). Replacing a W4 agent's action at the first decision where the two models' actions differ in tool structure (text-only differences are skipped) changes the outcome in 15.5–20.9% of task-block units, rescuing about as often as damaging; the net gain is -1.9 to $+2.9$ pp. This is a local effect, not end-to-end equivalence.
- **R2, where to replace** (§5). In about a quarter of W4 divergent states (pooled 26.9%), the swap's effect depends on who continues; strict reversals are rare and the mean interaction is near zero. Orderings of the three gains are exploratory: on 80 re-run states, $G - H$ is -2.5 pp under the original simulator and -8.5 pp under a second (change -6.0 [-16.2, $+4.2$]); no ordering is established that transfers.
- **R3, sampling and environment** (§6). At outcome-independent states most outcome changes under our simulator are text differences, but these labels did not reproduce reliably under another simulator. On 34 pre-registered unseen telecom tasks, the swap again rescues and damages, task fragility repeats, and the estimated effect is -4.6 pp (90% CI -7.2 to -2.0), inconclusive under the frozen readings. BFCL and ALFWorld also turn over both ways.
- **R4, finding replacements in advance** (§7). Q's token uncertainty is associated with divergences, and a frozen trigger captures more outcome-changing decisions than random selection at two of three rates, but is not shown to pick net-beneficial replacements.

Contributions. (1) An auditable same-state crossed intervention protocol for quantized agents: state restoration with bit-exact replay acceptance, full-action interventions, paired estimands (G, H, F and the $(\Delta_{\mathrm{Q}}, \Delta_{\mathrm{B}})$ table) with explicit denominators, and inference that separates a fixed task pool from new tasks. The 2×2 crossing itself is a standard factorial design. (2) An empirical analysis over weight, KV-cache and stress quantization, two model sizes and three benchmarks (R1–R4). Withdrawn results are in Appendix K.

2 Related Work

Episode-level comparisons of quantized agents. *Flat Score, Amplified Failures* (Jang et al., 2026) is the closest study in setting. On τ^2-bench retail and telecom it shows that 4-bit agents keep a flat task score while process damage grows. It reports a two-sided paired task split (33 tasks worse,

26 better for Gemma-4-31B on telecom) and explains the flat score by the error budget and by recovery, equally frequent at both precisions (55% after a failed call). Its comparisons are between independently sampled episodes, so the precision of the action that went wrong and the precision of the controller that must recover change together. Our design holds the state fixed and varies the two separately. H measures what the recovering controller contributes on identical histories, and the $(\Delta_{\mathrm{Q}}, \Delta_{\mathrm{B}})$ table measures whether an action's consequence depends on that controller. Two-sided flips under compression were first shown on static tasks (Dutta et al., 2024); agent-compression benchmarks report averages (Dong et al., 2025).

Live forks with a model switch. *The Replay Gap* (Gonuguntla, 2026) forks live SWE-agent trajectories at a fixed step and continues them with a different model (Qwen3-4B FP8 vs. Qwen3-14B AWQ), against same-model control forks. At the switch point the new model produces the next action *and* the continuation, so the two change together; this is our F contrast, not G or H. Its 359 control forks had 0 success flips; we claim no noise advantage.

Precision routing and its proxies. DMR (Li et al., 2026) learns a per-step BF16/W4 router from BF16-versus-quantized KL supervision and episode reward. Precision sensitivity has also been localised to phases (Lu et al., 2026) and task types (Zhang et al., 2026). We do not propose a competing router. We measure what a router would harvest, namely the intervention gains G, H and F at given states, and we test whether a Q-visible proxy predicts them. For quantized reasoning models, Lotfi et al. (2026) trace damage to high-entropy, high-KL token positions where markers such as "wait" are over-sampled; §7 tests the analogue for agent actions.

Evaluation noise and user simulators. Prefix caching changed multi-turn tool-use trajectories in 36.2% of episodes at 16-bit and 75.0% at 4-bit (Patodiya, 2026); agentic evaluations vary from run to run (Bjarnason et al., 2026; Chen et al., 2026), and with the user model (§6).

3 Crossed Same-State Intervention

3.1 Setting

Tasks and agents. We use τ^2-bench in text mode. The development panel has 58 retail and 27 airline tasks plus 80 telecom development tasks. A further 34 telecom tasks form a holdout that no agent ran before its protocol was frozen. The agents are Qwen3-4B-Instruct (q4) and Qwen3-8B with thinking disabled (q8). The user simulator is one greedy Qwen3-30B-A3B-Instruct model shared by all agents. Retail and airline success is the final-database match; telecom uses the official task-basis success. Without a user simulator, we also run BFCL multi-turn and ALFWorld, once per item (Appendix A).

Quantization and serving. Q is a simulated (fake-quantized) checkpoint served with BF16 compute on the same path as B. The main cells are GPTQ (Frantar et al., 2023) and AWQ (Lin et al., 2024) at W4 and a KV8 cache; stress cells are GPTQ-W3, RTN-W3 and a Q/K-rotated KV4 cache. B and Q run on vLLM with batch-invariant kernels and deterministic compilation, in separate server processes, so no prefix cache is shared across weights. A *block* is one closed-loop run of Q over the task pool under one prompt nonce, a random session-identifier line in the system prompt. Nonce blocks are the replicate unit.

3.2 Frozen states and estimands

Frozen state. Before Q's decision t in a recorded trajectory, we restore the state s_t. It comprises the full message history; the environment and user-device databases, rebuilt by strict replay of every recorded tool call; the user simulator's visible conversation; and the used step and error budget. B and Q each generate a full action from the same request, and the prompt token ids both servers receive must be identical. The state is written to disk before any branch runs.

Crossed intervention. $Y_{\mathrm{XY}} \in \{0, 1\}$ is task success when the action at s_t comes from X and controller Y takes over from the next step to the end, with X, Y $\in \{\mathrm{B}, \mathrm{Q}\}$. The action is executed by the real environment or answered by the real simulator and consumes one step; all four branches start from the same frozen state with the original remaining budget. Each escalation option is a paired contrast with QQ:

$$G = Y_{\mathrm{BQ}} - Y_{\mathrm{QQ}} \quad \text{(this step only)},$$
$$H = Y_{\mathrm{QB}} - Y_{\mathrm{QQ}} \quad \text{(from the next step)},$$
$$F = Y_{\mathrm{BB}} - Y_{\mathrm{QQ}} \quad \text{(from here on)}.$$

Their paired differences compare the options directly: $G - H = Y_{\mathrm{BQ}} - Y_{\mathrm{QB}}$, $F - G = Y_{\mathrm{BB}} - Y_{\mathrm{BQ}}$

and $F - H = Y_{\mathrm{BB}} - Y_{\mathrm{QB}}$. The effect of swapping the action can be read under either controller: $\Delta_{\mathrm{Q}} = Y_{\mathrm{BQ}} - Y_{\mathrm{QQ}}$ (Q continues; equal to G) and $\Delta_{\mathrm{B}} = Y_{\mathrm{BB}} - Y_{\mathrm{QB}}$ (B continues). Each takes values in $\{-1, 0, +1\}$, which gives a 3×3 table per state population. On its diagonal, the action swap has the same effect under both controllers. Off the diagonal, the interaction $I = \Delta_{\mathrm{Q}} - \Delta_{\mathrm{B}}$ is nonzero. The two corners $(+1, -1)$ and $(-1, +1)$ are *strict reversals*: the better action under one controller is the worse one under the other. These contrasts judge consequences, not correctness.

Rescue and damage. For the current-action swap, a *rescue* is $(Y_{\mathrm{BQ}}, Y_{\mathrm{QQ}}) = (1, 0)$ and a *damage* is $(0, 1)$. With rates r and d, the outcome turnover is $M = r + d$ and the net gain is $r - d$.

3.3 Replay acceptance

Causal reading requires exact replay. Before any development outcome was analysed, an engineering gate on real divergence states in retail, airline and telecom passed every check (one B→B field, `official_reward`, was missing from older trajectories and was recomputed from the stored reward information before re-verdicting). Q→Q no-ops matched in 20/20 states and B→B no-ops in 10/10, in every prompt-token hash, message, database hash, termination, reward and budget counter. In production, a sampled QQ no-op passed 924/924 units of the development analysis (snapshot 20261001T0600Z) and 328/328 sampled Bank U states over f01–f03; the Bank U gate passed a further 135/135 no-ops. Serving closes the prefix-cache channel (Patodiya, 2026).

3.4 State banks

Bank P: first tool-structure divergences. For each task, block and quantized cell, we walk along Q's trajectory to the first decision at which a_{B} and a_{Q} differ in kind (tool call, text, stop), tool name, call count or order, or normalised arguments. **Differences in text only are skipped**: two different natural-language replies, or identical calls with different accompanying text, are not Bank P states, and the walk continues past them.

Bank U: ordinary states. Bank U draws one decision per task uniformly, with the draw fixed by a hash, and its manifest was frozen before any generation. It includes *all* full language actions: every state runs all four branches unless a_{B} and a_{Q} are token-identical, and no effect is imputed. U1 states come from each quantized cell's own trajectories. U2B (BF16 trajectories) and U2Q (Q trajectories) give every cell of a model the identical state (a common-controller check is in Appendix F).

3.5 Denominators and inference

We use three denominators and never mix them. (1) *Full pool*: every task × block unit of a cell, with units that have no Bank P state counted as zero. r, d, M and the net gain $G_{\mathrm{all}} = r - d$ use it (§4). (2) *Divergent states*: the Bank P states of one cell, as a ratio of block sums. G, H, F and the 3×3 table use it (§5); G_{all} is G diluted by the share of units without divergence. (3) *Bank U*: one state per task, task-equal, per sub-bank (§6).

All tasks of a block share its nonce, and all branches share each state. Every within-cell contrast is therefore paired and *identified* for that cell's state population. Comparisons across cells (q4 vs. q8, AWQ vs. GPTQ, W4 vs. KV8) compare different state populations, because each cell selects its own first divergences, so they are *descriptive*. Only complete blocks enter; a block with any missing unit leaves the analysis. The primary interval treats the task pool as fixed: a block bootstrap with at least 10 complete blocks, otherwise a t interval over block means (stress cells have 3 blocks). A crossed task × block bootstrap is the interval for generalising to new tasks. More nonce blocks narrow only the first, and we never let it stand in for the second. Bank U is analysed over blocks f01–f03; its primary intervals are t over block means.

4 R1: What Replacing the Current Action Rescues and Damages

Two-sided turnover, small net gain. On the τ^2-bench panel (Table 1; 85 tasks × 20 complete blocks), replacing a W4 agent's first tool-structure-divergent action changes the task outcome in $M = 15.5$–20.9% of units. Rescues and damages are of similar size, $r = 6.8$–11.4 and $d = 7.5$–10.1 pp, so the net gain G_{all} lies between -1.9 and $+2.9$ pp. Two fixed-pool intervals exclude zero, negative for q4 GPTQ-W4 and positive for q8 GPTQ-W4; none of the new-task intervals does. Pooled over the four W4 cells, G_{all} is $+0.7$ $[-0.3, +1.6]$ pp with $M = 18.6\%$. KV8 cells turn over fewer outcomes ($M = 6.5$–8.1%; a descriptive comparison across cells), and four of the five stress cells reach W4-like turnover ($M = 18.4$–20.4%) with only three blocks; q8 RTN-W3, whose own success is

Cell	units	div.	r	d	M	G_{all} [fixed pool]	[new tasks]
q4 GPTQ-W4	85×20	1102	6.8	8.7	15.5	−1.9 [−3.7, −0.1]	[−6.8, +2.7]
q4 AWQ-W4	85×20	1439	9.8	10.1	19.9	−0.3 [−2.2, +1.6]	[−5.4, +5.2]
q8 GPTQ-W4	85×20	1473	10.4	7.5	17.9	+2.9 [+1.4, +4.4]	[−2.9, +9.2]
q8 AWQ-W4	85×20	1562	11.4	9.5	20.9	+1.9 [−0.6, +4.3]	[−3.9, +8.1]
W4 pool	85×20		9.6	9.0	18.6	+0.7 [−0.3, +1.6]	
q4 KV8	85×22	593	3.1	3.4	6.5	−0.3 [−1.7, +1.0]	[−3.0, +2.3]
q8 KV8	85×22	855	4.3	3.7	8.1	+0.6 [−0.4, +1.6]	[−1.9, +3.0]
q4 GPTQ-W3^s	85×3	232	9.0	10.2	19.2	−1.2 [−7.0, +4.7]	[−8.6, +5.9]
q4 KV4-rot^s	85×3	233	11.8	7.1	18.8	+4.7 [−4.1, +13.5]	[−3.1, +12.9]
q8 GPTQ-W3^s	85×3	242	10.2	8.2	18.4	+2.0 [−12.5, +16.4]	[−6.3, +11.8]
q8 KV4-rot^s	85×3	239	11.8	8.6	20.4	+3.1 [−14.5, +20.8]	[−5.9, +12.9]
q8 RTN-W3^s	85×3	255	3.1	3.5	6.7	−0.4 [−3.8, +3.0]	[−4.7, +3.9]

Table 1: Replacing Q's first divergent action while Q continues, τ^2-bench retail/airline panel (pp). Denominator: the full pool of tasks × complete blocks; units without a Bank P state count as zero. div. = Bank P states. $G_{\text{all}} = r - d$ is a local replacement effect, *not* the end-to-end success difference (Appendix B). Fixed pool: block bootstrap (≥10 blocks) or t over block means; new tasks: crossed task×block bootstrap. ^{s}stress cell, 3 blocks. Source: `analysis/fork/out/dev_all_20261001T070326Z.md`.

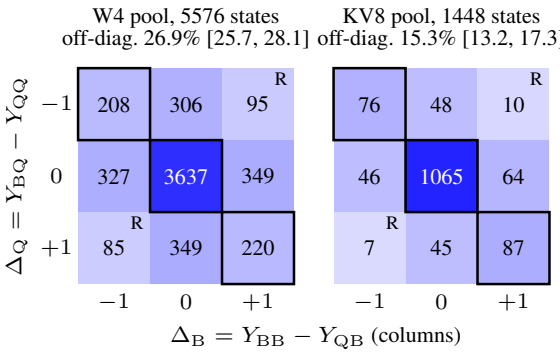


Figure 2: Does the action swap have the same effect under both controllers? Counts of (Δ_Q, Δ_B) over Bank P states, summed over the W4 cells (left) and KV8 cells (right). Boxed: diagonal. R: strict reversals. Shading: share of states (power scale). Cell counts are summed over cells (summed off-diagonal 27.1% and 15.2%); headers give equal-weight cell averages with fixed-pool 95% intervals. A marginal-preserving pairing agreement control (states re-paired within a cell, both marginals kept) expects 38.0% (W4) and 33.6% (KV8) off-diagonal; $\kappa = 0.29$ and 0.55. Sources: `analysis/fork/out/ghf_20261001T211519Z.md` (§2), `analysis/fork/out/agreement_control_20261002T024240Z.md`.

low, turns over 6.7%. A small net gain thus coexists with substantial two-sided turnover. G_{all} is not an end-to-end success comparison (Appendix B).

Turnover concentrates in tasks (exploratory). The top 20% of tasks hold 57–71% of all flips in the W4 and KV8 cells, and task-level turnover and its direction repeat across disjoint halves of the nonce blocks. This is a retrospective association on viewed development data (Appendix G). On unseen telecom tasks, the pre-registered repeatability test (S2) passed: per-task turnover in one half of the nonce blocks predicts turnover in the other half better than persona-preserving random pairing ($\rho = 0.59$, $p = .0001$; §6). S2 does not test concentration or direction, and it does not rule out task-difficulty explanations.

5 R2: Where to Replace

On average over W4 cells, no option has a detectable gain. Pooled over the four W4 cells, $G = +0.5$, $H = +0.2$ and $F = +0.7$ pp, and every interval covers zero, as do their differences.

Within cells, the paired gains differ. In q8 GPTQ-W4, $G = +3.3$ [+1.6, +5.0] pp exceeds $H = +0.4$ ($G - H = +2.9$ [+1.3, +4.7]), and $F = +5.4$ exceeds G ($F - G = +2.1$ [+0.2, +4.0]). In q8 AWQ-W4, $G - H = +3.4$ [+1.5, +5.2] and $F - G = -1.5$ [−4.2, +0.8]. In q4 GPTQ-W4, $G = -2.9$ and $G - H = -3.6$ [−6.6, −0.6]. In the stress cell q8 RTN-W3 (three blocks, exploratory), $H = +15.3$ and $F = +18.4$ against $G = -0.4$ ($G - H = -15.7$ [−27.5, −3.9]). In q4 KV8, B continuing after Q's action lowers success: $H = -2.5$ [−4.5, −0.5] (fixed pool, unadjusted; exploratory). These intervals treat the 85 tasks as fixed. Holm-adjusted over the four core W4 cells, $G - H$ remains below .05 in both q8 cells ($p = .013$, .012) but not in q4 GPTQ-W4 ($p = .069$), and for new tasks the $G - H$ intervals of all four W4 cells cover zero (Table 5). We therefore treat the $G - H$ results as exploratory; a smaller second-simulator control does not confirm them (§6). Cells select different states, so orderings across cells are descriptive and cannot be attributed to the quantizer or bit width alone.

A BF16 reference action can lower success. In q4 GPTQ-W4, B's action lowers success under Q's continuation on the fixed task pool ($G = -2.9$ [−5.7, −0.2] pp, unadjusted; new-task interval covers zero), with a similar point estimate under B's ($\Delta_B = -3.2$ [−6.4, +0.2]); the reference action itself is not reliably better here.

The action's consequence often depends on who continues. Figure 2 shows the (Δ_Q, Δ_B) table. In 26.9% [25.7, 28.1] of pooled W4 states, the action swap has a different effect under B's continuation than under Q's; per W4 cell the share

Cell	states	G (this step)	H (next step on)	F (from here on)	$G - H$	$F - G$	$F - H$ (= Δ_B)	off-diag. %	rev. %	mean I
q4 GPTQ-W4	1102	−2.9 [−5.7, −0.2]	+0.7 [−1.6, +3.0]	−2.5 [−5.3, +0.2]	−3.6 [−6.6, −0.6]	+0.5 [−1.2, +2.2]	−3.2 [−6.4, +0.2]	25.0	2.6	+0.3
q4 AWQ-W4	1439	−0.3 [−2.6, +1.9]	+0.9 [−1.1, +2.9]	−0.8 [−3.3, +1.6]	−1.3 [−3.0, +0.6]	−0.4 [−2.4, +1.5]	−1.7 [−3.6, +0.4]	29.3	3.1	+1.3
q8 GPTQ-W4	1473	+3.3 [+1.6, +5.0]	+0.4 [−1.2, +1.9]	+5.4 [+3.5, +7.2]	+2.9 [+1.3, +4.7]	+2.1 [+0.2, +4.0]	+5.0 [+3.2, +6.8]	22.9	2.7	−1.7
q8 AWQ-W4	1562	+2.1 [−0.6, +4.7]	−1.3 [−3.5, +0.9]	+0.6 [−1.5, +2.7]	+3.4 [+1.5, +5.2]	−1.5 [−4.2, +0.8]	+1.9 [−0.1, +3.8]	30.5	4.2	+0.3
W4 pool	5576	+0.5 [−0.7, +1.7]	+0.2 [−1.3, +1.6]	+0.7 [−0.6, +2.1]	+0.4 [−0.8, +1.7]	+0.2 [−1.0, +1.2]	+0.5 [−1.0, +2.1]	26.9	3.2	+0.0
q4 KV8	593	−1.0 [−5.3, +3.2]	−2.5 [−4.5, −0.5]	−1.2 [−4.7, +2.3]	+1.5 [−3.1, +6.0]	−0.2 [−3.5, +3.7]	+1.3 [−2.5, +4.9]	15.7	1.2	−2.4
q8 KV8	855	+1.3 [−0.9, +3.4]	−0.4 [−2.1, +1.5]	+2.5 [+0.0, +5.1]	+1.6 [−0.6, +3.9]	+1.2 [−0.5, +3.2]	+2.8 [+0.3, +5.5]	14.9	1.2	−1.5
KV8 pool	1448	+0.1 [−2.8, +3.0]	−1.4 [−2.7, −0.1]	+0.6 [−1.8, +3.1]	+1.6 [−1.2, +4.1]	+0.5 [−1.6, +2.8]	+2.1 [−0.2, +4.2]	15.3	1.2	−1.9
q4 GPTQ-W3^s	232	−1.3 [−7.7, +5.0]	+4.3 [−10.7, +19.4]	+1.3 [−1.9, +4.4]	−5.6 [−15.0, +3.7]	+2.6 [−0.6, +5.8]	−3.0 [−15.1, +9.1]	31.5	3.9	+1.7
q4 KV4-rot^s	233	+5.2 [−4.5, +14.9]	+9.4 [+5.1, +13.7]	+12.4 [+11.0, +13.9]	−4.3 [−18.3, +9.8]	+7.3 [−2.0, +16.5]	+3.0 [−1.9, +8.0]	33.0	1.7	+2.1
q8 GPTQ-W3^s	242	+2.1 [−13.3, +17.2]	+0.4 [−7.4, +8.2]	+2.1 [−7.4, +11.4]	+1.7 [−9.1, +12.3]	+0.0 [−11.0, +11.1]	+1.7 [−0.1, +3.4]	23.1	4.5	+0.4
q8 KV4-rot^s	239	+3.3 [−15.4, +21.9]	−0.8 [−18.2, +16.3]	+4.6 [−16.0, +25.0]	+4.2 [+2.5, +5.8]	+1.3 [−1.8, +4.3]	+5.4 [+0.8, +10.0]	31.0	4.6	−2.1
q8 RTN-W3^s	255	−0.4 [−3.8, +3.0]	+15.3 [+0.7, +29.9]	+18.4 [+16.7, +20.1]	−15.7 [−27.5, −3.9]	+18.8 [+13.8, +23.9]	+3.1 [−13.0, +19.2]	29.0	0.4	−3.5

Table 2: Three replacement gains and their paired differences on each cell's Bank P states (pp; same denominator in every column). $G = Y_{BQ} - Y_{QQ}$, $H = Y_{QB} - Y_{QQ}$, $F = Y_{BB} - Y_{QQ}$. Off-diag.: share of states with $\Delta_Q \neq \Delta_B$; rev.: strict reversals. Fixed-pool 95% intervals (block bootstrap over 20–22 complete blocks; t over 3 blocks for ^{s}stress cells, which cannot be pooled). New-task intervals are in Table 5. Within a row, contrasts are paired and identified; between rows, descriptive. Source: `analysis/fork/out/ghf_20261001T211519Z.md`.

is 22.9–30.5%. In the KV8 pool it is 15.3% [13.2, 17.3]; the W4-KV8 difference is descriptive. A marginal-preserving pairing agreement control, which re-pairs states at random within a cell, expects 38.0% [36.7, 39.0] in the W4 pool ($\kappa = 0.29$ [0.27, 0.32], new-task [0.22, 0.36]) and 33.6% in KV8 ($\kappa = 0.55$; Appendix D.1): the effects under the two controllers agree more than random cross-state pairing would. This control is not a no-interaction null. At the same state, in our deterministic, replay-verified setting, $I \neq 0$ is a non-additive interaction on the success scale, whose identification does not depend on exceeding the control; it does not by itself establish a recovery or quantization-specific mechanism, or a deployment gain. Strict sign reversals are rare: 3.2% of pooled W4 states, and 1.2–4.2% of states in each W4 and KV8 cell. They make up 8–14% of the off-diagonal mass. The rest is an effect under one controller and none under the other.

Why the mean interaction is near zero. $E[I]$ is the difference between the swap's average effect under the two controllers, whereas the off-diagonal share is $P(I \neq 0)$. States where the swap helps more under Q ($I > 0$) and under B ($I < 0$) are about equally frequent (13.6% and 13.4% in the W4 pool), so $E[I] = +0.0$ [−1.9, +2.0] pp although 26.9% of states depend on the controller (new-task interval 22.1–32.0%). The value of a fixed replacement policy is $E[G]$, $E[H]$ or $E[F]$, not $E[I]$. All 16 outcome patterns: Appendix C.

6 R3: Do Sampling and Environment Change the Answer?

Ordinary states turn over less, mostly through text under our simulator. Descriptively, at ordinary states replacing Q's action changes the outcome less often than at first tool-structure divergences (different state populations; no interval on the difference). Over blocks f01–f03 (task-equal; t intervals over block means), M per sub-bank is 7.8–9.0% for the q4 W4 cells in U1 and U2B, 4.7–9.0% for the q8 W4 cells and at most 4.7% for KV8. The net gain lies within ±2.8 pp (U1, U2B) and ±5.0 pp (U2Q), and every interval covers zero (Appendix F). Identical actions never change the outcome, as determinism implies. After deduplication, 150 of the 208 outcome-changing cell×state rows (≈72.1%) are class (ii); the rows are not independent states: the three blocks repeat the same tasks, and each U2B/U2Q state appears once per cell of a model (Appendix Table 10): the same tool structure with different text, nearly always two natural-language replies. 26 involve a different tool name, call count or argument. This is a *composition share*: per instance, text, tool/argument and act/talk differences flip the outcome in 14.4%, 18.2% and 18.9% of rows. Text dominates because it is common (1045 of 1357 differing rows), not because each text difference matters more. Among simple content features, only an above-median length difference between the two texts co-occurs with outcome change (in all four U1/U2B × model groups); question, confirmation-request and new-entity flags show no consistent association. These are untested regular-expression features (Appendix F). Bank P cannot see these differences by design.

Sensitivity to the user simulator. The user simulator is a plausible mediator of text consequences; changing it moves τ-bench outcomes by 9–15 points (Seshadri et al., 2026; Jain and Sandhu, 2026). We froze a control before running it: the 39 deduplicated class-(ii) flips of block f01 and

39 matched non-flipping states. Their BQ and QQ branches were each re-run three times with a different user simulator, Sonnet 5.5 answering through a file relay from Claude Code subagents. Judged by the sign of the replicate mean, only 7 of the 39 flips keep their direction (95% CI 0.075–0.335). Six reverse, and 26 vanish (15 both succeed, 2 both fail, 9 ties). Among the matched non-flip states, 11 of 39 become flips (0.15–0.45). Replicates of the same arm disagree in 41% of flip arms and in 21–26% of control arms. **The Bank U text-difference outcome labels did not reproduce reliably under this alternative relay setting.** This is not evidence that text consequences are simulator artefacts: the control changes the simulator model, adds sampling randomness and uses a relay interface (scope and sensitivity analyses in Appendix F).

Sensitivity of the four-arm results. Before running, we sampled 80 Bank P states without outcome information (20 per W4 cell, stratified by domain and block) and re-ran all four arms five times each with Sonnet 5.5 as the user, one isolated call per user turn (7896 calls; all integrity checks passed). Per-state arm outcomes track the original single runs ($r = 0.63$–0.68). Pooled over cells, $G = -3.0$ [-10.2, $+4.8$], $H = +5.5$ [$+0.0$, $+11.2$] and $G - H = -8.5$ [-14.8, -2.8] pp under Sonnet, against $G - H = -2.5$ [-11.2, $+6.2$] on the same states under the original simulator (change -6.0 [-16.2, $+4.2$]). This sample is too small to test the few-pp $G - H$ contrasts of Table 2; in it, even the original simulator gives $G - H = 0.0$ for both q8 W4 cells. Both pooled $G-H$ estimates are negative and their change covers zero, so R2's within-cell orderings are neither confirmed nor shown to reverse under another simulator. An exploratory mean squared interaction under Sonnet is in Appendix D.2.

Unseen tasks: telecom holdout. Before any holdout unit was run, we froze a confirmatory protocol (Appendix J): 34 unseen telecom tasks $\times$ 12 fresh nonce blocks $\times$ 3 cells (q4 GPTQ-W4, q8 GPTQ-W4, q8 AWQ-W4, weighted 1/2, 1/4, 1/4); primary endpoint G_{all} on a crossed 90% interval against ± 3 pp; secondaries two-sidedness and repeatability of task fragility. All 12 blocks were complete; 1200 of the 1224 units were analysed, after 2 structural task$\times$cell exclusions (Appendix J). The primary endpoint is $G_{\text{all}} = -4.57$ pp, crossed 90% interval [-7.17, -1.96] (block-t [-6.08, -3.05]). The pre-registered readings define only a positive gain, a gain above 3 pp and a small effect, so the decisional reading is *inconclusive*. Descriptively (this is not one of the pre-registered readings), the interval lies entirely below zero: on these unseen tasks, replacing the first divergent action with BF16's lowered average success. Both secondaries pass. Rescues and damages are each above zero ($r = +5.21$ [$+3.30$, $+7.11$], $d = +9.77$ [$+6.45$, $+13.10$] pp), and task fragility repeats across independent nonce halves ($\rho = 0.59$, permutation $p = .0001$). *Post hoc* (descriptive, outside the confirmation; Table 17), R2 on the 1180 divergent states gives $G = -4.6$ [-7.8, -1.5], $H = -2.8$ [-6.3, $+0.6$], $F = -2.5$ [-6.2, $+1.3$], $G - H = -1.8$ [-4.8, $+1.1$] and mean $I = -5.0$ [-9.6, -0.4] pp (crossed 95%), unlike the near-zero development W4 pool, and 20.7% off-diagonal, against 25.7% under the marginal-preserving pairing agreement control ($\kappa = 0.19$; not a no-interaction null). This G conditions on divergence and nearly equals G_{all} because 1180 of 1200 units diverge. The holdout confirms, as pre-specified, two-sided consequences (S1) and repeatable task fragility (S2). Its primary endpoint is inconclusive under the frozen readings. Descriptively, its estimate is negative. It does not confirm any other development result.

No user simulator: BFCL and ALFWorld. BFCL multi-turn and ALFWorld have no simulated user; each item runs once, and intervals are item-level. Replacing a W4 agent's first divergent action changes the outcome in 5.2–24.6% of items, in both directions, so two-sided turnover also occurs without a simulated user (ALFWorld: q8 only). The net gain covers zero except for BFCL q4 AWQ-W4, where it is -3.8 [-6.5, -1.3] pp. Controller dependence also appears without an LLM user (exploratory; Appendix D.3): pooled off-diagonal shares are 10.3% (BFCL W4) and 36.4% (q8 ALFWorld W4).

7 R4: Can Beneficial Replacements Be Found in Advance?

G, H and F are oracle quantities that require running B and all four branches; a deployable rule sees only Q. Following Lotfi et al. (2026), we scored every Bank U f01 state at the token level (1140 cell $\times$ state rows, 58 flips), teacher-forcing both models on the generated ids of a_{B} and a_{Q}. We record truncated top-20 entropy, the top-1 mar-

gin and a top-20-union KL, not full entropy or KL. All analyses are exploratory, and f01 is development data.

Uncertainty is associated with divergence. Q's entropy at the start of its action separates states where B and Q act differently from states where they act identically, with AUC 0.66–0.91 per sub-bank and model. Over all ordinary states, flips also sit at lower Q margins (U1: AUC 0.31 [0.22, 0.40] for q4, 0.24 [0.13, 0.36] for q8; stratified).

Among divergent states, it does not reliably identify consequences. Restricting to states where the actions differ removes the "any difference at all" signal. In U1 the signal is not detectable: no Q-visible statistic has an interval excluding 0.5 (21 and 9 flips; f01). Only small sub-banks keep a signal: q4 U2B (mean entropy over a_{Q}, AUC 0.73 [0.53, 0.89]) and q4 U2Q (entropy and margin at the action start, AUC 0.90 and 0.10; 10 states, 4 flips). At the first divergent token, pooled entropy and KL do not separate flips from non-flips except in q4 U2Q (KL AUC 0.80 [0.65, 0.93], entropy 0.25 [0.07, 0.48]; 4 flips). Per cell, two further unadjusted intervals exclude 0.5, in opposite directions (U1 q4 GPTQ-W4 entropy 0.30; U2B q4 KV8 KL 0.10). No class of first divergent token reproduces the marker enrichment of Lotfi et al. (2026). The largest q4 enrichment is ordinary word choice (rate ratio 1.89, Fisher $p = .019$, not corrected). On development data, uncertainty is associated with *where* quantization changes the action more strongly than with *which* changes matter.

A frozen trigger, tested on held-out blocks. We froze a Q-visible trigger on f01 before any f02/f03 state was scored: escalate one decision when the minimum Q margin over a_{Q} is low, with per-cell thresholds at 5, 10 and 20% escalation rates (rule hash in Appendix H). On held-out f02/f03 (1020 U1 rows, 69 turnover states) it escalated 50, 114 and 222 states. Net gains were $+4$, $+2$ and $+5$ (rescues/damages 6/2, 7/5, 16/11) against matched-count random expectations of $+0.21$, $+0.98$ and $+2.38$ (exact one-sided hypergeometric $p = .0345, .417, .262$). It captured 8, 12 and 27 turnover states against 3.48, 7.52 and 14.30 expected ($p = .0182, .0613, .0002$). With no level fixed in advance, net gain passes a nominal one-sided .05 only at the 5% rate, and not after Holm correction. At 20% (21.8% of rows selected), the net gain of about 0.49 pp exceeds random (0.23 pp) by 0.26 pp. The frozen Q-only rule has not been shown to select net-beneficial replacements, and two new nonce blocks of the same tasks do not establish routing gains on new tasks.

Headroom. A hindsight action-replacement oracle with the same per-cell counts nets $+3.33$, $+3.73$ and $+3.73$ pp; the rule captures 4/34, 2/38 and 5/38 of that net. A menu oracle and a post-hoc $\mathrm{KL}(\mathrm{B}\|\mathrm{Q})$ row (not Q-only; no significant net gain, Holm $p = 1.0$) are in Appendix H.

8 Discussion

When to escalate. Average success, action divergence and replacement gain answer different questions and can disagree. The four W4 cells have small end-to-end differences with wide intervals (Appendix B). Replacing their first tool-structure-divergent action turns over 15.5–20.9% of task×block units (full pool), but the pooled net gain is near zero ($+0.7$ [-0.3, $+1.6$] pp). Reporting only the first two cannot tell whether a BF16 fallback would raise success. Text actions should be evaluated as actions but, given §6, not reported as a property of the action without a simulator control. We recommend reporting r, d and M next to any net gain, each with its denominator, and keeping fixed-pool and new-task intervals apart.

Which part to replace. We found no replacement unit that is best in every cell. On the fixed task pool, $G - H$ is positive in the two q8 W4 cells (Holm $p = .013, .012$) and strongly negative in the severe q8 RTN-W3 stress cell. These within-cell orderings are exploratory: new-task intervals mostly cover zero, a second simulator (W4 cells only) neither confirms nor reverses the q8 orderings, and cross-cell contrasts are descriptive because cells select different states. Measure G, H and F per configuration and environment.

Escalation can hurt. Replacements damage as well as rescue (§4). In q4 KV8, B continuing has a negative fixed-pool estimate ($H = -2.5$ [-4.5, -0.5] pp, unadjusted; new-task interval covers zero; §5); on 34 unseen telecom tasks, replacing the first divergent action has an estimated effect of -4.6 pp (crossed 90% CI -7.2 to -2.0; pre-registered primary reading inconclusive; §6).

Limitations

What we do not claim. We do not claim success equivalence between BF16 and W4 agents; equivalence tests at ± 3 pp on new-task intervals fail. G, H and F are oracle quantities, not the value of a deployable policy (§7). We do not claim a W$\times$KV interaction, and we do not present static precision restoration as a method (Appendix K).

One model family, two sizes. All agents are Qwen3 models (4B and 8B). Effects may differ at other scales and in other families. End-to-end runs exist for larger Qwen models, but they have no crossed forks, and we draw no conclusion from them (Appendix E).

Simulated quantization. Quantized checkpoints are dequantized and served with BF16 compute on a deterministic path. Deployed low-bit kernels change the numerics and add serving variation that we deliberately remove. We make no claims about latency, memory or deployed kernels, and the cost of a BF16 fallback is outside our scope.

One user simulator; two small controls. All τ^2-bench episodes use a single greedy simulator, and continuation effects include its replies. Both second-simulator controls replace only the user after a fixed history, use a stochastic Sonnet 5.5 user (a subagent relay for Bank U, isolated single calls for Bank P), cover 78 and 80 states, and say nothing about real users. The Bank P control has 20 states per cell, too few to confirm or refute few-pp contrasts. We did not run a same-precision perturbation 2×2 (e.g. B versus B under another nonce), so the off-diagonal share is not calibrated against controller-independent sensitivity: in this deterministic setting it includes sensitivity of the continuation to any change of controller.

LLM annotation. Process labels in Appendix I come from LLM annotators (Sonnet 5.5, two independent instances plus an adjudicator) and rule-based checks. No human annotation has been done. Agreement among instances of one model family shows reproducibility, not correctness.

Small confirmation and partial banks. The pre-registered holdout covers 34 telecom tasks and three cells. Its negative estimate is established only for telecom, these three cells and this simulator, and its pre-specified readings did not include a negative-gain category. Bank P intervenes only at the first tool-structure divergence per task and block. Bank U samples one state per task and block and has three blocks, so its block intervals rest on two degrees of freedom. Prompt nonces define replicate blocks; they are a researcher-chosen perturbation, not a deployment distribution.

Ethics Statement

This work involves no crowdworkers and no personal data. All users are simulated by an LLM on public benchmark tasks, and all agents are openly released Qwen3 checkpoints. The process annotations were produced by LLMs on synthetic benchmark conversations. The main risk we see is overreading. A small forked-state gain on one simulator and a fixed task pool does not certify a quantized agent for deployment, and our end-to-end comparisons do not establish equivalence. Deployment decisions in consequential settings should not rest on these measurements alone.

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A Setup Details

Splits. The development split has 58 retail, 27 airline and 80 telecom tasks. The holdout has 34 telecom tasks, and its pre-registration hash is recorded in the split file. The telecom split was fixed from τ^2-bench's 114-task base split by a stratified, salted hash before any run. The external benchmarks use BFCL multi-turn (base, miss_func, miss_param and long_context, 197 items each) and ALFWorld (134 out-of-distribution games). Items were selected by hash in manifests committed before any model run.

Success endpoints. For retail and airline we use the final-database match with normal termination. This is not τ^2-bench's official reward, which additionally scores natural-language or communicate checks. Telecom uses the official deterministic task-basis success. Context overflow and unparseable output count as failures. Infrastructure outages are missing, never failures, and a block with a missing unit leaves the primary analysis.

Fork mechanics. The forced action is returned by a fork agent for the one step that follows h_t. A tool call is executed by the real environment, and text is answered by the real simulator, which never reuses observations from the original trajectory. tau2 resets step and error counters when it starts from a history. The fork restores them to the values used at t, so the remaining budget equals the original. Every max_steps and too_many_errors episode on the checked blocks is reproduced by the recount. Fork state files record a stale default `user_model_id`; the served simulator is identified from the replica registry.

B End-to-End Success Comparison

The replacement gains G_{all}, G, H and F are not end-to-end comparisons. Table 3 gives the end-to-end comparison, computed separately on the same checkpoints, the same tasks and the same 20 nonce blocks as the forks.

Cell	Q / BF16 (%)	Q−BF16 [95% CI]	NI 3 pp	TOST ±3 pp
q4 GPTQ-W4	40.0 / 35.8	+4.2 [−2.4, +10.5]	pass	fail
q4 AWQ-W4	36.8 / 35.8	+0.9 [−5.8, +7.5]	fail	fail
q8 GPTQ-W4	30.1 / 30.4	−0.2 [−7.7, +6.8]	fail	fail
q8 AWQ-W4	31.5 / 30.4	+1.2 [−5.3, +7.6]	fail	fail
pool, all four		+1.5 [−2.7, +5.6]	pass	fail

Table 3: End-to-end closed-loop success, quantized minus BF16 (pp). Units are paired on nonce, and the crossed task×block bootstrap is drawn jointly across cells. Non-inferiority (NI) at a 3 pp margin passes for q4 GPTQ-W4 and the all-four pool. Equivalence (TOST at ±3 pp) passes nowhere. These comparisons therefore support neither a difference nor equivalence. Source: `analysis/insights/iter_4/out/success_ni_ALIGNED_20260930T213009Z.md`.

C All 16 Outcome Patterns and New-Task Intervals

Table 4 lists every four-branch pattern per cell; `0000` (39–58%) and 1111 (6–20%) dominate. Table 5 gives the new-task intervals for Table 2.

$(Y_{\mathrm{BB}}, Y_{\mathrm{BQ}}, Y_{\mathrm{QB}}, Y_{\mathrm{QQ}})$	q4 AWQ	q4 GPTQ-W3	q4 GPTQ	q4 KV4r	q4 KV8	q8 AWQ	q8 GPTQ-W3	q8 GPTQ	q8 KV4r	q8 KV8	q8 RTN-W3
0000	634 (44.1%)	105 (45.3%)	472 (42.8%)	92 (39.5%)	286 (48.2%)	767 (49.1%)	130 (53.7%)	766 (52.0%)	113 (47.3%)	497 (58.1%)	141 (55.3%)
0001	55 (3.8%)	7 (3.0%)	29 (2.6%)	3 (1.3%)	14 (2.4%)	66 (4.2%)	2 (0.8%)	36 (2.4%)	8 (3.3%)	15 (1.8%)	3 (1.2%)
0010	57 (4.0%)	13 (5.6%)	54 (4.9%)	13 (5.6%)	11 (1.9%)	38 (2.4%)	8 (3.3%)	34 (2.3%)	10 (4.2%)	15 (1.8%)	20 (7.8%)
0011	58 (4.0%)	6 (2.6%)	71 (6.4%)	4 (1.7%)	37 (6.2%)	30 (1.9%)	9 (3.7%)	49 (3.3%)	3 (1.3%)	39 (4.6%)	0 (0.0%)
0100	50 (3.5%)	6 (2.6%)	36 (3.3%)	9 (3.9%)	8 (1.3%)	53 (3.4%)	8 (3.3%)	45 (3.1%)	13 (5.4%)	20 (2.3%)	1 (0.4%)
0101	21 (1.5%)	4 (1.7%)	17 (1.5%)	4 (1.7%)	11 (1.9%)	36 (2.3%)	7 (2.9%)	12 (0.8%)	5 (2.1%)	5 (0.6%)	2 (0.8%)
0110	22 (1.5%)	5 (2.2%)	14 (1.3%)	2 (0.9%)	2 (0.3%)	33 (2.1%)	4 (1.7%)	16 (1.1%)	2 (0.8%)	5 (0.6%)	1 (0.4%)
0111	41 (2.8%)	5 (2.2%)	21 (1.9%)	7 (3.0%)	13 (2.2%)	51 (3.3%)	6 (2.5%)	31 (2.1%)	7 (2.9%)	7 (0.8%)	6 (2.4%)
1000	51 (3.5%)	8 (3.4%)	32 (2.9%)	19 (8.2%)	16 (2.7%)	67 (4.3%)	10 (4.1%)	75 (5.1%)	16 (6.7%)	28 (3.3%)	32 (12.5%)
1001	23 (1.6%)	4 (1.7%)	15 (1.4%)	2 (0.9%)	5 (0.8%)	33 (2.1%)	7 (2.9%)	24 (1.6%)	9 (3.8%)	5 (0.6%)	0 (0.0%)
1010	18 (1.3%)	5 (2.2%)	13 (1.2%)	9 (3.9%)	4 (0.7%)	17 (1.1%)	5 (2.1%)	17 (1.2%)	3 (1.3%)	3 (0.4%)	20 (7.8%)
1011	36 (2.5%)	9 (3.9%)	33 (3.0%)	9 (3.9%)	8 (1.3%)	32 (2.0%)	3 (1.2%)	19 (1.3%)	2 (0.8%)	11 (1.3%)	6 (2.4%)
1100	44 (3.1%)	3 (1.3%)	51 (4.6%)	9 (3.9%)	41 (6.9%)	43 (2.8%)	10 (4.1%)	82 (5.6%)	9 (3.8%)	46 (5.4%)	2 (0.8%)
1101	36 (2.5%)	7 (3.0%)	27 (2.5%)	3 (1.3%)	9 (1.5%)	38 (2.4%)	4 (1.7%)	23 (1.6%)	1 (0.4%)	11 (1.3%)	1 (0.4%)
1110	51 (3.5%)	9 (3.9%)	15 (1.4%)	10 (4.3%)	7 (1.2%)	65 (4.2%)	4 (1.7%)	34 (2.3%)	6 (2.5%)	10 (1.2%)	4 (1.6%)
1111	242 (16.8%)	36 (15.5%)	202 (18.3%)	38 (16.3%)	121 (20.4%)	193 (12.4%)	25 (10.3%)	210 (14.3%)	32 (13.4%)	138 (16.1%)	16 (6.3%)
n	1439	232	1102	233	593	1562	242	1473	239	855	255

Table 4: All 16 four-branch outcome patterns per cell: count (share of the cell's Bank P states). Pattern digits are $(Y_{\mathrm{BB}}, Y_{\mathrm{BQ}}, Y_{\mathrm{QB}}, Y_{\mathrm{QQ}})$. KV4r: Q/K-rotated KV4; W3 and KV4r cells are stress cells with 3 blocks. Source: `analysis/fork/out/ghf_20261001T211519Z.md` (§3).

Cell	G	H	F	$G-H$	$F-G$	$F-H$
q4 GPTQ-W4	$[-10.3, +4.2]$	$[-4.0, +6.0]$	$[-9.9, +4.7]$	$[-11.2, +3.6]$	$[-4.7, +6.1]$	$[-11.1, +4.4]$
q4 AWQ-W4	$[-6.5, +6.1]$	$[-3.7, +5.5]$	$[-7.4, +5.8]$	$[-7.7, +5.1]$	$[-6.2, +5.4]$	$[-8.7, +4.7]$
q8 GPTQ-W4	$[-3.3, +10.5]$	$[-3.8, +4.9]$	$[-2.0, +13.0]$	$[-3.2, +9.3]$	$[-3.0, +7.1]$	$[-2.0, +12.5]$
q8 AWQ-W4	$[-4.1, +8.7]$	$[-7.1, +4.6]$	$[-6.2, +7.3]$	$[-1.6, +8.6]$	$[-7.6, +4.2]$	$[-3.8, +7.5]$
W4 pool	$[-2.9, +3.8]$	$[-2.7, +2.8]$	$[-3.1, +4.4]$	$[-3.1, +3.7]$	$[-2.8, +3.1]$	$[-3.5, +4.5]$
q4 KV8	$[-9.4, +7.3]$	$[-6.4, +1.1]$	$[-9.3, +7.7]$	$[-7.1, +9.7]$	$[-5.7, +5.3]$	$[-6.7, +9.5]$
q8 KV8	$[-4.0, +6.9]$	$[-3.5, +2.9]$	$[-3.5, +8.1]$	$[-3.8, +7.2]$	$[-2.5, +5.0]$	$[-3.1, +8.7]$
KV8 pool	$[-5.3, +5.6]$	$[-3.7, +0.9]$	$[-4.5, +5.8]$	$[-3.8, +6.6]$	$[-2.8, +3.9]$	$[-3.0, +7.3]$
q4 GPTQ-W3^s	$[-9.7, +6.6]$	$[-6.6, +14.5]$	$[-6.8, +9.4]$	$[-15.0, +4.0]$	$[-5.2, +10.5]$	$[-12.2, +6.6]$
q4 KV4-rot^s	$[-3.4, +14.2]$	$[+2.1, +17.0]$	$[+3.0, +21.6]$	$[-13.5, +7.0]$	$[-1.3, +17.1]$	$[-6.2, +12.2]$
q8 GPTQ-W3^s	$[-6.7, +12.1]$	$[-7.2, +8.1]$	$[-6.6, +11.8]$	$[-7.7, +11.2]$	$[-9.1, +9.1]$	$[-6.8, +11.0]$
q8 KV4-rot^s	$[-6.6, +13.8]$	$[-11.2, +8.5]$	$[-6.6, +15.9]$	$[-3.3, +12.2]$	$[-7.2, +9.9]$	$[-2.5, +14.6]$
q8 RTN-W3^s	$[-4.7, +3.9]$	$[+5.9, +25.9]$	$[+9.4, +28.2]$	$[-24.7, -7.1]$	$[+10.2, +27.8]$	$[-7.1, +13.3]$

Table 5: New-task generalisation: crossed task×block bootstrap 95% intervals for the estimates of Table 2 (pp; same Bank P states). These answer whether the effect would hold for new tasks drawn like the panel; the fixed-pool intervals of Table 2 treat the 85 tasks as fixed. ^{s}stress cell, 3 blocks. Source: `analysis/fork/out/ghf_20261001T211519Z.md` (§1).

D Supplementary R2 Analyses

D.1 Marginal-preserving pairing agreement control

For each cell, observed agreement is the diagonal share of the $(\Delta_{\mathrm{Q}}, \Delta_{\mathrm{B}})$ table, $p_o = \sum_k P(\Delta_{\mathrm{Q}} = k, \Delta_{\mathrm{B}} = k)$, so the off-diagonal share is $1 - p_o = P(I \neq 0)$. The control re-pairs states independently within a cell while keeping both marginals, $p_e = \sum_k p_Q(k)\, p_B(k)$, and Cohen's $\kappa = (p_o - p_e)/(1 - p_e)$. Pools use the paper's equal weight per cell over common complete blocks, with the same bootstrap draws as Table 2. This is a pairing agreement control, not a no-interaction null and not a chance baseline for controller dependence (§5).

D.2 Mean squared interaction under the second simulator (exploratory)

On the 80 SIMCTRL-P states (§6), each arm has $n_a = 5$ replicates under Sonnet 5.5. Per state, $\hat{I}_s = \bar{Y}_{\mathrm{BQ}} + \bar{Y}_{\mathrm{QB}} - \bar{Y}_{\mathrm{BB}} - \bar{Y}_{\mathrm{QQ}}$ and $U_s = \hat{I}_s^2 - \sum_a \bar{Y}_a(1 - \bar{Y}_a)/(n_a - 1)$, which is unbiased for I_s^2 if, given the state, the replicates of each arm are i.i.d. Bernoulli and the four arms are independent (no shared cache or seed). Pooled with equal weight over the four W4 cells, mean $\hat{I}_s^2$ is 0.110, the noise correction 0.060 and mean U_s 0.050, with 95% intervals [0.004, 0.107] over states and [0.001, 0.111] over task clusters. One cell is negative (q4 AWQ-W4, -0.025) and is kept, as are negative per-state values. The estimate is the mean squared probability-scale interaction over these 80 frozen states. It is not the share of states with interaction and not a variance, it is not comparable with the deterministic 26.9% of §5, and we make no significance claim. Source: `analysis/fork/out/simctrl_p_interaction_20261002T024357Z.md`.

D.3 R2 estimands on BFCL and ALFWorld (supplementary, exploratory)

Table 7 applies R2 to the external benchmarks, from the raw four-arm outcomes on divergent items (one run per item; item bootstrap within

Cell	states	off-diag. %	exp. %	κ [fixed pool]	κ [new tasks]
q4 GPTQ-W4	1102	25.0 [22.8, 27.3]	40.5	0.381 [0.325, 0.437]	[0.245, 0.520]
q4 AWQ-W4	1439	29.3 [27.2, 31.2]	38.5	0.238 [0.215, 0.261]	[0.138, 0.343]
q8 GPTQ-W4	1473	22.9 [21.1, 24.5]	36.3	0.369 [0.324, 0.409]	[0.235, 0.492]
q8 AWQ-W4	1562	30.5 [28.3, 32.6]	36.8	0.171 [0.128, 0.213]	[0.075, 0.274]
W4 pool	5576	26.9 [25.7, 28.1]	38.0	0.291 [0.265, 0.318]	[0.223, 0.359]
q4 KV8	593	15.7 [12.7, 18.6]	36.2	0.567 [0.487, 0.643]	[0.412, 0.705]
q8 KV8	855	14.9 [12.9, 17.0]	31.1	0.522 [0.462, 0.584]	[0.386, 0.656]
KV8 pool	1448	15.3 [13.2, 17.3]	33.6	0.546 [0.487, 0.604]	[0.436, 0.652]
q4 GPTQ-W3^s	232	31.5 [10.5, 52.6]	36.1	0.129 [−0.258, 0.534]	[−0.049, 0.347]
q4 KV4-rot^s	233	33.0 [17.6, 48.3]	38.0	0.131 [−0.016, 0.280]	[−0.034, 0.298]
q8 GPTQ-W3^s	242	23.1 [6.0, 40.2]	36.4	0.364 [0.190, 0.548]	[0.199, 0.532]
q8 KV4-rot^s	239	31.0 [19.0, 43.0]	37.7	0.179 [0.098, 0.251]	[0.032, 0.327]
q8 RTN-W3^s	255	29.0 [24.6, 33.5]	28.6	−0.016 [−0.233, 0.200]	[−0.101, 0.142]

Table 6: Pairing agreement control per cell (Bank P divergent states). Off-diag.: observed share with $\Delta_{\mathrm{Q}} \neq \Delta_{\mathrm{B}}$ (fixed-pool 95%); exp.: share expected under marginal-preserving within-cell re-pairing. Pools: equal weight per cell. ^{s}Stress cells (3 blocks; t intervals; flagged, never pooled). In q8 RTN-W3, $\Delta_{\mathrm{Q}} = 0$ on 93.3% of states, so p_e is close to p_o and $\kappa \approx 0$. Source: `analysis/fork/out/agreement_control_20261002T024240Z.md`.

benchmark category). It is post hoc and not part of any pre-registered analysis. In the KV4 cells $Y_{\mathrm{QQ}} = 0$ on every state, so Δ_{Q} is almost constant, their off-diagonal mass is H and κ is near 0 by construction. In BFCL q4 GPTQ-W3, the base category was dropped whole by the QQ no-op rule, leaving 591 items and 540 states. Off-diagonal shares are 4.7–17.1% in BFCL W4 cells (pool 10.3%) and 32.8–40.0% in q8 ALFWorld W4 (pool 36.4%), with pairing κ 0.44–0.71 and 0.10–0.24 (incl. GPTQ-W3). The ALFWorld GPTQ-W3 stress cell has $H = +38.5$ and $G - H = -34.6$ $[-43.5, -25.4]$ pp; KV4 cells reduce to H.

E Coverage Matrix

Table 8 lists the completed crossed-intervention coverage behind every result, from the W93 audit. It counts tasks, states and complete blocks after the strict admission rule; queued or running work is not counted.

F Bank U per Sub-Bank

Table 9 gives the per-state turnover behind §6, over blocks f01–f03; it supersedes the single-block f01 table of an earlier draft. Exact action agreement is 0.43–0.83 per cell and sub-bank.

Text content. Over f01–f03, an above-median relative length difference between the two texts co-occurs with outcome change in U1 and U2B of both models (flipping pairs among above- vs. below-median pairs: q4 U1 30/154 vs. 15/155; q8 U1 22/101 vs. 6/101; q4 U2B 33/138 vs. 11/139; q8 U2B 16/99 vs. 9/100). In f01 alone this was visible only for q4. Question, confirmation-request and new-entity flags show no consistent association. The features are regular expressions, and this is descriptive.

Bank U simulator control: sensitivity and scope. Dropping two mechanically flagged replies leaves the reading of §6 unchanged (6, 6 and 25 of 37; Table 11). In either direction, 13 of the 39 original flip states flip again, against 11 of 39 new flips at matched non-flip states; this does not show that the two groups are equivalent. The relay's responders may have seen other requests. The control's scope is narrow: fixed histories from the original simulator, a stochastic relay rather than API Sonnet or real users, block f01 only and three replicates.

Shared fragility across quantizers (a lead). On the 255 shared q4 U2B states of f01–f03, cell pairs co-flip 21 times, against about 3.3 expected under independence, and 16 of the 21 co-flips go in the same direction. They come from 15 distinct states, 3 of which flip under all three cells. On q8 U2B, where every f01 flip was single-cell, 10 states now flip under two or more cells (14 pairwise co-flips against about 2.8 expected; 10 of 14 in the same direction). Joint flips almost never share an identical quantized action across cells (0 of 21 in q4, 2 of 14 in q8). We still read this as a lead toward *state-level* fragility, where different actions at the same state flip the outcome, and not yet as a finding: the co-flips are counted on one fixed set of states per block, with no correction.

A common controller (Q0). On the frozen 25% subsample of U2 in f01–f03, also continued by a common controller Q0, B's action leads to the identical outcome in every cell on all 146 shared states, which checks determinism. Turnover under Q0 is not consistently above or below the native turnover on the same rows (q4 AWQ-W4 U2B 8/58 vs. 5/58; q8 AWQ-W4 U2B 4/52 vs. 5/52; q4 KV8 U2B 1/58 vs. 2/58), so the f01 observation that Q0 turnover is at least the native one does not hold over three blocks. With 17–58 states per cell and sub-bank, this is too small to say whether the controller changes which actions matter.

G Concentration of Turnover in Tasks

This appendix completes the exploratory concentration result of §4 (development data, 85 tasks ×

Cell	states	G	H	F	$G - H$	off-diag. %	exp. %	κ	rev. (n)
BFCL q4 GPTQ-W4	573	+2.8 [−0.4, +5.9]	+1.4 [−0.7, +3.5]	+4.2 [+0.9, +7.2]	+1.4 [−1.8, +4.5]	11.7	27.4	0.574	3
BFCL q4 AWQ-W4	642	−4.7 [−7.8, −1.5]	−1.1 [−3.6, +1.4]	−1.4 [−4.8, +1.7]	−3.6 [−6.7, −0.2]	17.1	32.4	0.471	8
BFCL q8 GPTQ-W4	533	+0.9 [−1.3, +3.3]	+1.5 [+0.0, +3.2]	+2.1 [−0.4, +4.6]	−0.6 [−3.1, +2.0]	4.7	16.2	0.710	1
BFCL q8 AWQ-W4	548	+2.0 [−0.4, +4.7]	+1.8 [+0.0, +3.7]	+3.6 [+0.9, +6.5]	+0.2 [−2.6, +3.1]	7.8	19.9	0.607	0
BFCL W4 pool		+0.3 [−1.2, +1.8]	+0.9 [−0.2, +2.0]	+2.1 [+0.3, +3.9]	−0.6 [−2.3, +1.0]	10.3			
ALFWorld q8 GPTQ-W4	115	−4.3 [−13.3, +5.0]	+7.0 [−2.6, +16.7]	+9.6 [−0.9, +19.7]	−11.3 [−20.9, −1.8]	40.0	46.4	0.137	4
ALFWorld q8 AWQ-W4	119	−4.2 [−13.6, +4.9]	+5.0 [−2.6, +12.2]	+4.2 [−5.9, +13.7]	−9.2 [−18.8, +0.0]	32.8	42.9	0.237	3
ALFWorld W4 pool		−4.3 [−11.1, +2.2]	+6.0 [−0.5, +12.6]	+6.9 [−2.1, +15.5]	−10.3 [−17.7, −2.9]	36.4			
BFCL q4 GPTQ-W3^s	540	+2.8 [+0.2, +5.5]	+5.2 [+2.9, +7.7]	+8.9 [+5.7, +12.2]	−2.4 [−5.7, +0.7]	13.0	23.3	0.443	1
BFCL q8 GPTQ-W3^s	696	+1.3 [−1.1, +3.7]	+1.4 [−0.1, +2.9]	+3.4 [+0.9, +6.0]	−0.1 [−2.6, +2.4]	8.3	20.4	0.592	1
BFCL q4 KV4^s	787	+0.3 [+0.0, +0.6]	+2.9 [+1.9, +4.1]	+20.3 [+17.6, +23.0]	−2.7 [−3.9, −1.5]	20.5	20.9	0.020	0
BFCL q8 KV4^s	788	+0.3 [+0.0, +0.6]	+1.8 [+1.0, +2.7]	+14.5 [+12.1, +17.0]	−1.5 [−2.5, −0.6]	14.7	15.2	0.029	0
ALFWorld q8 GPTQ-W3^s	130	+3.8 [−1.5, +9.8]	+38.5 [+29.8, +46.9]	+42.3 [+33.6, +50.8]	−34.6 [−43.5, −25.4]	30.8	34.1	0.098	0
ALFWorld q8 KV4^s	134	+0.0 [+0.0, +0.0]	+38.1 [+30.6, +45.5]	+53.0 [+44.8, +61.2]	−38.1 [−45.5, −30.6]	26.9	26.9	0.000	0

Table 7: R2 estimands on BFCL multi-turn and ALFWorld (supplementary, exploratory, post hoc; success scale, no LLM user simulator; pp, divergent items; 95% item-bootstrap intervals). Off-diag.: share with $\Delta_{\mathrm{Q}} \neq \Delta_{\mathrm{B}}$; exp. and κ: marginal-preserving pairing agreement control; rev.: strict reversals. Pools: equal weight per cell. ^{s}Stress and unrotated KV4 cells; in KV4 $Y_{\mathrm{QQ}} = 0$ on every state, so the table reduces to H. Source: `analysis/fork/out/ext_r2_20261002T024433Z.md`.

Cell	τ^2-bench retail (Bank P)	τ^2-bench airline (Bank P)	telecom dev (Bank P)	Bank U (retail / airline)	BFCL / ALFWorld
q4 GPTQ-W4	58 t · 727 st · 20/20 b	27 t · 375 st · 20/20 b	holdout dirs exist (unread)	3 b · 393 / 177 rows	788 items · 573 st / –
q4 AWQ-W4	58 t · 986 st · 20/20 b	27 t · 453 st · 20/20 b	–	3 b · 393 / 177 rows	788 items · 642 st / –
q4 KV8	58 t · 412 st · 22/22 b	27 t · 181 st · 22/22 b	–	3 b · 393 / 177 rows	– / –
q8 GPTQ-W4	58 t · 1017 st · 20/20 b	27 t · 456 st · 20/20 b	80 t · 477 st · 6/6 b; holdout dirs	3 b · 393 / 177 rows	788 · 533 st / 134 · 115 st
q8 AWQ-W4	58 t · 1087 st · 20/20 b	27 t · 475 st · 20/20 b	80 t · 480 st · 6/6 b; holdout dirs	3 b · 393 / 177 rows	788 · 548 st / 134 · 119 st
q8 KV8	58 t · 585 st · 22/22 b	27 t · 270 st · 22/22 b	–	3 b · 393 / 177 rows	– / –
q4 GPTQ-W3	58 t · 162 st · 3/3 b	27 t · 70 st · 3/3 b	–	–	591 · 540 st† / –
q4 KV4-rot	58 t · 164 st · 3/3 b	27 t · 69 st · 3/3 b	–	–	– / –
q4 RTN-W3	0 t · 0 st · 0/3 b	0 t · 0 st · 0/3 b	–	–	– / –
q8 GPTQ-W3	58 t · 166 st · 3/3 b	27 t · 76 st · 3/3 b	–	–	788 · 696 st / 134 · 130 st
q8 KV4-rot	58 t · 165 st · 3/3 b	27 t · 74 st · 3/3 b	–	–	– / –
q8 RTN-W3	58 t · 174 st · 3/3 b	27 t · 81 st · 3/3 b	–	–	– / –
q4 / q8 KV4 (unrotated)	–	–	–	–	788 · 787 / 788 · 788 st; q8 ALFWorld 134 · 134 st

Table 8: Completed crossed-intervention coverage (t = tasks, st = states, b = complete/found blocks; BFCL and ALFWorld: items · states, one run). Bank U: blocks completed and state rows per domain; blocks f01–f03 are analysed (q8 KV8 f03 finished after the coverage snapshot; Appendix F). "holdout dirs": telecom holdout fork directories exist and were not opened. The q14, q32 and l8 checkpoints of the source matrix have end-to-end runs only, and q4 RTN-W3 has no complete Bank P block. The telecom development analysis of §6 uses the planning fit over these 6 blocks. Source: `analysis/fork/out/ghf_20261001T211519Z.md` (§5). †The BFCL base category of q4 GPTQ-W3 was dropped by the QQ no-op rule (Appendix D.3). The end-to-end q4 AWQ-W3 run is incomplete: two airline episodes hit 600 s serving timeouts (infrastructure), so it is not reported; its completed episodes and failure logs are kept.

20 blocks). The per-task dispersion of M is 4.0–8.7 times its binomial null in the W4 and KV8 cells, and the top 20% of tasks hold 57–71% of all flips. We split the 20 blocks into two disjoint halves of 10 nonces. Task-level M correlates across halves (Spearman 0.52–0.86). Tasks that were net-rescue in one half have a mean net gain in the other half that is 14–51 pp higher than that of net-damage tasks, in all 12 cell-directions. A Freedman–Lane test conditions on measured baselines: QQ success propensity and BF16 success, plus trajectory length in a post-hoc variant. With them, one half's task rate still predicts the other's in all 6 configurations × 2 directions for both M and G (Holm-adjusted). This is a *retrospective conditional association* on viewed development data. It is carried by retail. Airline alone gives 8 of 12 for M (5 after Holm), and 3 of 12 under a heteroscedasticity-robust wild bootstrap. Direction persistence beyond flip opportunity holds more often for rescues than for damages, in 11 versus 6 of the 12 cell-directions. Different quantizers flip the same tasks: the task-identity null is rejected in 7 of 8 quantizer pairs. When both flip, however, they agree in direction only 42–81% of the time. This overlaps with the stable failure set of Jang et al. (2026). What is new here is the signed, two-sided view: shared fragility, but with near-random relative direction across quantizers.

H Trigger Headroom and a KL Reference Row

Table 12 compares the frozen trigger of §7 (rule sha256 `335e737d. . .`, unchanged) with two references on the same 1020 f02/f03 U1 rows, which hold 38 rescues and 31 damages. The *matched*

Cell	U1 (n=255)	U2B (n=255)	U2Q (n=60)
q4 GPTQ-W4	.090 (11/12)	.082 (10/11)	.083 (4/1)*
q4 AWQ-W4	.082 (7/14)	.078 (7/13)	.083 (3/2)
q4 KV8	.047 (4/8)	.039 (5/5)	.017 (0/1)
q8 GPTQ-W4	.047 (8/4)	.067 (9/8)	.050 (3/0)*
q8 AWQ-W4	.086 (14/8)	.090 (15/8)	.067 (3/1)
q8 KV8	.035 (5/4)	.031 (6/2)	0 (0/0)

Table 9: Bank U, blocks f01–f03: per-state turnover M (rescues/damages), task-equal (one state per task and block; 85/85/20 states per block). G lies within ± 0.028 in U1 and U2B and within ± 0.050 in U2Q, and every interval includes 0. Intervals are t intervals over the three block means and are wide (e.g. q4 GPTQ-W4 U1: .090 [.029, .151]); all are in the source. *GPTQ-W4's U2Q states are its own U1 states, with concordant outcomes on 20/20 duplicates per block; they are counted once in cross-sub-bank totals. Source: `analysis/fork/out/bank_u_f01_f02_f03_20261002T003037Z.md`.

Model:sub	(i) identical	(ii) same tool structure, text	(iii) different tool or args	(iv) act vs. talk
q4:U1	399: 0	309: 45 (.146)	19: 1 (.053)	38: 10 (.263)
q8:U1	483: 0	202: 28 (.139)	47: 11 (.234)	33: 4 (.121)
q4:U2B	430: 0	277: 44 (.159)	24: 4 (.167)	34: 3 (.088)
q8:U2B	470: 0	199: 25 (.126)	40: 9 (.225)	56: 14 (.250)
q4:U2Q	108: 0	63: 10 (.159)*	5: 0	4: 1 (.250)
q8:U2Q	129: 0	33: 6 (.182)*	11: 1 (.091)	7: 0
unique rows	1943: 0	1045: 150 (.144)	143: 26 (.182)	169: 32 (.189)

Table 10: Which kind of action difference changes the outcome at ordinary states (Bank U, blocks f01–f03). Entries: "state rows: outcome-changing rows (flip rate)", pooled over the three cells of a model within a sub-bank. The last row counts each (cell, state) once: q4/q8 GPTQ-W4's U2Q states repeat their U1 states (* includes 5 and 3 repeated class-(ii) flips), so 216 flipping rows become 208. In block f01 alone the corresponding counts were 39 of 56 flips in class (ii), with rates .126/.109/.200 for (ii)/(iii)/(iv). Source: `analysis/fork/out/bank_u_f01_f02_f03_20261002T003037Z.md`; f01: `analysis/fork/out/bank_u_f01_20261001T134456Z.md`.

Analysis	flips	retained	rev.	vanished (s/f/tie)	ctrl	new flips (resc./dam.)
All replies	39	7 [0.07, 0.34]	6	26 (15/2/9)	39	11 [0.15, 0.45] (7/4)
Flagged dropped	37	6 [0.06, 0.32]	6	25 (15/2/8)	39	11 [0.15, 0.45] (7/4)

Table 11: Bank U simulator control (block f01; Sonnet 5.5 relay user, $R = 3$). Fates use the sign of the replicate-mean flip indicator; s/f: both arms succeed/fail. "Flagged dropped": the two arm replicates with a mechanically flagged reply are set to missing. Replicates of one arm disagree in 16/39 (BQ) and 16/39 (QQ) flip states and in 10/39 and 8/39 control states. Per single replicate, retained flips are 5, 7 and 4 of 39. Source: `analysis/fork/out/simctrl_20261001T212742Z.md`, `analysis/fork/out/simctrl_relay_flagged_20261001T225847Z.md`.

action-replacement oracle selects, per cell, exactly as many rows as the frozen rule did, taking only rows with $G > 0$ and filling with $G = 0$ rows when a cell has too few rescues; it never selects a damage. It is hindsight selection, not a policy. At the 10% and 20% budgets every rescue fits. The *three-option menu oracle* $E[\max(G, H, F, 0)]$ is the share of states where some arm beats QQ. On Bank P divergent states it is 0.202 over the four W4 cells (state-weighted) against 0.117 for the action alone (1127 vs. 654 of 5576 states), and 0.191 vs. 0.113 over the six core cells. It is hindsight selection that also allows different replacement types, with different costs, so it is not a same-cost deployment comparison and not comparable with the action rule.

The *KL reference row* is a post-hoc auxiliary baseline. Its score is a truncated $\mathrm{KL}(\mathrm{B}\|\mathrm{Q})$ over the union of both models' top-20 next-token sets plus a tail bucket, per a_{Q} token, teacher-forced on identical prompt ids. It needs a BF16 prefill of the prompt and a_{Q} at every decision (here 6709.3 prompt and 82.3 action tokens on average), so it is not Q-only. It captures turnover above chance (33 vs. 16.7 at 20%) but no significant net gain (Holm $p = 1.0$). Three aggregations were fit on f01 only, with per-cell thresholds and the frozen rule's selection criterion; KLu_max was chosen, and all of f02/f03 is reported. The two candidates not chosen on f01 look better on f02/f03 (e.g. KLu_mean at 20%: net +9, Holm $p = .0852$), but selecting them now would be selection on validation data. Because the f02/f03 outcomes had been seen, the row is post hoc, and it is not a DMR reproduction, which trains a router.

I Controlled Injection and Process Audit

A natural explanation of R1 is a local-defect story: the quantized agent makes a local error, and the task fails unless the continuation recovers. Two analyses constrain this explanation. They limit what it can explain; they do not identify a mechanism.

Controlled defect injection. We froze 30 ordinary states from q4 GPTQ-W4 trajectories, all from one development block. At each state we forced one of three actions as the next step: an INValid action, a matched VALid action, or a process-perturbation control (PP) that repeats the last successful read. Either B or Q then continued. The defect categories were wrong_id, wrong_tool

Budget	same per-cell selection counts sel.	oracle r/d/net (pp)	frozen r/d/net (pp)	captured	KL reference (KLu_max; needs BF16; post hoc) sel.	net vs. random (p; Holm)	r/d	turnover vs. random (p)
5%	50	34/0/+34 (+3.33)	6/2/+4 (+0.39)	4/34	72	−1 vs. +0.45 (.8337; 1.0000)	5/6	11 vs. 4.58 (.0038)
10%	114	38/0/+38 (+3.73)	7/5/+2 (+0.20)	2/38	126	+1 vs. +0.82 (.5483; 1.0000)	8/7	15 vs. 7.54 (.0048)
20%	222	38/0/+38 (+3.73)	16/11/+5 (+0.49)	5/38	257	+3 vs. +1.59 (.3984; 1.0000)	18/15	33 vs. 16.73 (.0000)

Table 12: Frozen Q-only trigger against a matched action-replacement oracle (hindsight; same per-cell selection counts) and a post-hoc KL reference row, on f02/f03 U1 (1020 rows). Captured: frozen net over oracle net. Random: matched-count per-cell random-trigger expectation; p: exact one-sided hypergeometric, Holm over the three budgets. Source: `analysis/fork/out/oracle_trigger_20261002T024423Z.md`.

Ctrl	Contrast	resc./dam.	turnover	net [95% CI]
B	INV vs VAL	4/3	7/30	+0.03 [−0.13, +0.20]
B	PP vs VAL	3/2	5/30	+0.03 [−0.10, +0.20]
B	INV vs PP	2/2	4/30	0.00 [−0.13, +0.13]
Q	INV vs VAL	2/2	4/30	0.00 [−0.13, +0.13]
Q	PP vs VAL	3/2	5/30	+0.03 [−0.10, +0.17]
Q	INV vs PP	2/3	5/30	−0.03 [−0.20, +0.10]
Controller dependence D ($\neq 0$ in 8/30 states)				+0.03 [−0.17, +0.27]

Table 13: Defect injection on 30 frozen q4 states (development, one block): per-state rescue/damage of one arm against another, under the BF16 (B) or quantized (Q) continuation controller. $D = [Y(\text{INV}, B) - Y(\text{VAL}, B)] - [Y(\text{INV}, Q) - Y(\text{VAL}, Q)]$. Task-clustered bootstrap intervals; all exact McNemar $p \geq 0.25$. Source: `analysis/recovery/inject_q4/summary_v2.md`.

and premature_write, with 10 states each. Every state passed a replay audit of 22 recorded fields (VAL×Q against the original episode). Three fields were not recorded: the counters right after the restore, the rendered-prompt hash sequence and the tool-schema hash sequence.

Result: two-sided turnover, no clear error-specific difference detected. Injected defects change the outcome in 7/30 states under B (4 rescues, 3 damages) and 4/30 under Q (2 and 2). The process-perturbation control changes 5/30 under both controllers (Table 13). The error-specific effect, INV minus PP, is 0.00 [−0.13, +0.13] under B and −0.03 [−0.20, +0.10] under Q. The controller dependence D is nonzero in 8 of 30 states, but its mean is +0.03 [−0.17, +0.27]. So injected defects cause heterogeneous two-sided turnover, and at $n = 30$ states we detect no clear difference from a generic process perturbation and no mean controller difference. The intervals are wide, so this does not show that errors and generic perturbations are equivalent. An earlier reading of the same data, that injected defects "seldom change outcomes", was wrong and has been retracted (Appendix K).

Manipulation checks limit the categories. Two of 30 states failed a manipulation check: an injected write was rejected, and a valid reference itself returned "not found". Restricting to the 28 passing states leaves the reading unchanged. The wrong_tool category mostly produced argument errors: 8 of 10 injected calls failed with an unexpected-keyword error. We therefore call it a *tool-call/schema error perturbation*, not a wrong-tool test. premature_write is mostly failure→failure: the valid arm under Q already fails in 7/10 of these states, and turnover is 0/10 under both controllers. The event chain error → correction → related requirement satisfied completes in only 6 (B) and 7 (Q) of 29 injected errors, although 17 are corrected under each controller. Injections add tool errors (+0.43 under B, +0.57 under Q per branch).

Replication on q8 GPTQ-W4. The same design on 30 frozen q8 states (block q3 of the GPTQ-W4 run; 10 per category) gives the same reading (Table 14). All 30 states pass the replay audit. Injected defects change the outcome in 5/30 states under each controller (3 rescues, 2 damages), the process-perturbation control in 6/30 (B) and 5/30 (Q). The error-specific effect is −0.03 [−0.13, +0.07] under B and −0.13 [−0.30, +0.03] under Q; the latter interval still allows a sizeable error-specific harm, and D is nonzero in 6 of 30 states with mean 0.00 [−0.17, +0.17]. Under Q, the process-perturbation control itself rescues 5 states and damages none (+0.17 [+0.03, +0.30], McNemar $p = 0.0625$). Manipulation checks fail more often than in q4: 7 of the 10 premature writes failed the check (6 were rejected by the environment, 1 executed without changing the database), and 2 valid references returned "not found". On the 21 passing states, INV vs VAL turnover is 5 (B) and 4 (Q), and $D \neq 0$ in 5. The q8 premature_write category therefore mostly did not inject a write.

Ctrl	Contrast	resc./dam.	turnover	net [95% CI]
B	INV vs VAL	3/2	5/30	$+0.03$ $[-0.10, +0.17]$
B	PP vs VAL	4/2	6/30	$+0.07$ $[-0.07, +0.23]$
B	INV vs PP	1/2	3/30	-0.03 $[-0.13, +0.07]$
Q	INV vs VAL	3/2	5/30	$+0.03$ $[-0.10, +0.17]$
Q	PP vs VAL	5/0	5/30	$+0.17$ $[+0.03, +0.30]$
Q	INV vs PP	2/6	8/30	-0.13 $[-0.30, +0.03]$
Controller dependence D ($\neq 0$ in 6/30 states)				0.00 $[-0.17, +0.17]$

Table 14: Defect injection on 30 frozen q8 GPTQ-W4 states (development, block q3), as in Table 13. Task-clustered bootstrap intervals; exact McNemar $p \geq 0.0625$. Source: `analysis/recovery/inject_q8/summary_v2.md`.

Process audit (LLM-annotated). We sampled 80 development fork states, 20 per outcome sign. Two independent Sonnet 5.5 instances annotated them blind, a third instance adjudicated, and rule-based programmatic labels were added. These annotations are a *process description* conditional on outcome-defined groups. They are not causal evidence, not prevalences and not human labels. Three patterns stand out (Table 15). First, local defects and final failure do not correspond one-to-one. In damage states, where Q's action led to success and B's to failure, Q's first-action defect is corrected later in 10 of 20 QQ branches. Second, whether a write executes does not separate the failing from the succeeding continuation of the same state (rescue 17 vs. 17; damage 15 vs. 14). Third, the failing continuation more often carries the rubric label `unintended_or_unconsented_change` (rescue: QQ 9 vs. BQ 2, paired difference $+0.35$ $[+0.08, +0.57]$; damage: 1 vs. 9, -0.40 $[-0.62, -0.11]$). In damage states, 7 of these 20 BQ branches carry the label on writes *with* recorded consent. That is the semantic "wrong change" part of the label, and it overlaps with the database-defined outcome by construction. Loops and budget exhaustion also mark failing branches (rescue: QQ 5 vs. BQ 0). We make no claim about consent. The programmatic consent field agrees poorly with the annotators, so we use it only as a mechanical row. Re-labelling 30 states with fresh instances of the same model family reproduces the key pattern. That shows reproducibility, not correctness.

I.1 Process audit (Table 4 v2)

The core "local defect" combines groundedness, argument validity and permission/consent judgements. The tool-validity field (inter-annotator $\kappa = 0.29$) is excluded from it. Mechanical fields agree with the programmatic verifier (write executed vs. the rubric's irreversible-change field, $\kappa = 0.99$). The consent fields agree only moderately ($\kappa = 0.44$ on writes both see).

J Holdout Pre-Registration

The protocol `docs/insights/PREREG_FORK_CONFIRM.md` was frozen before any holdout unit was run. Its sha256 is `902892dd63e99e16ab9beefee9d39319d835eacdada3d4d92c4a1022d13fd622`. It fixes the following design:

- tasks, cells, weights and nonce blocks;
- halves $H_A = \{h01, h04, h05, h09, h11, h12\}$ and H_B, the rest of $h01$–$h12$;
- an outcome-blind repair rule, with reserve blocks used only for infrastructure repair;
- the decisional crossed interval (two-way ANOVA moments, normal 90%);
- the readings: positive if $L > 0$, above 3 pp if $L > 3$, small if the interval lies inside ± 3 pp, otherwise inconclusive;
- the secondary families S1 (both $r > 0$ and $d > 0$) and S2 (persona-stratified permutation, 10,000 draws, no outcome screen);
- six stratum families, with Holm correction within each.

The analysis script refuses holdout data unless the hash matches. Two task-cell pairs were excluded as structural, and the complete-case analysis is identical. Table 16 gives the confirmatory result. Strata (six families, Holm within family) are descriptive here. Every level estimate is ≤ 0, and the Holm-significant levels are negative: step-budget T2 and T3, mobile-data issues (-9.09 pp) and the fewest-faults tercile (-6.37 pp).

Post-hoc R2 estimands on the holdout. Table 17 is a supplementary, descriptive output computed after the confirmatory analysis, with the same admission and units. It is not part of the pre-registered confirmation and changes neither the primary endpoint nor its reading. Its G is a ratio over divergent states, whereas G_{all} in Table 16 averages over all admitted units with non-divergent units counted as 0.

Label (source)	Comparison	rescue	damage	both succeed	both fail
unintended_or_unconsented_change (LLM)	QQ vs BQ	9 vs 2, $+0.35$ [$+0.08$, $+0.57$]	1 vs 9, -0.40 [-0.62, -0.11]	1 vs 1	5 vs 7, -0.10 [-0.35, $+0.17$]
write executed (programmatic)	QQ vs BQ	17 vs 17, 0.00 [-0.25, $+0.25$]	15 vs 14, $+0.05$ [-0.26, $+0.35$]	10 vs 9	11 vs 13
executed write w/o recorded consent (prog.)	QQ vs BQ	9 vs 5, $+0.20$ [-0.06, $+0.43$]	3 vs 3, 0.00 [-0.21, $+0.21$]	3 vs 2	4 vs 3
loop or budget exhaustion (LLM)	QQ vs BQ	5 vs 0, $+0.25$ [$+0.04$, $+0.47$]	0 vs 2, -0.10 [-0.30, $+0.08$]	1 vs 2	7 vs 4
action judged indefensible (LLM)	Q vs B action	7 vs 8, -0.05 [-0.37, $+0.28$]	12 vs 5, $+0.35$ [-0.05, $+0.65$]	10 vs 6	12 vs 9
local defect, core (LLM)	Q vs B action	4 vs 6, -0.10 [-0.33, $+0.14$]	12 vs 7, $+0.25$ [-0.09, $+0.53$]	6 vs 3	11 vs 7
QQ first-action defect corrected later (LLM)	single branch	4/20 [0.08, 0.42]	10/20 [0.30, 0.70]	4/20	7/20

Table 15: Process audit of 80 development fork states, 20 per outcome sign (rescue: $Y_{\mathrm{BQ}} = 1, Y_{\mathrm{QQ}} = 0$; damage: $Y_{\mathrm{BQ}} = 0, Y_{\mathrm{QQ}} = 1$). **LLM-annotated** by Sonnet 5.5 (two independent blind instances and an adjudicating instance), plus rule-based programmatic labels; no human labels. Counts are first vs. second of $n = 20$ states. Intervals are within-state paired differences with Newcombe 95% CIs; single-branch rows use Wilson 95% intervals. This is a process description conditional on outcome-defined groups: not causal, not prevalence. The label name is the rubric category; we make no claim about consent. Source: `analysis/annotation/wp5/table4_v2.md`.

Cell (weight)	units	excl.	G	r	d	ρ_c
q4 GPTQ-W4 (1/2)	396	1	-3.03 [-7.86, $+1.80$]	$+5.05$	$+8.08$	0.668
q8 GPTQ-W4 (1/4)	408	0	-5.64 [-9.33, -1.94]	$+4.41$	$+10.05$	0.374
q8 AWQ-W4 (1/4)	396	1	-6.57 [-11.98, -1.15]	$+6.31$	$+12.88$	0.644
Weighted (G_{all})			-4.57 [-7.17, -1.96]	$+5.21$	$+9.77$	0.588

Table 16: Telecom holdout, pre-registered analysis (34 tasks $\times$ 12 blocks; pp; crossed 90% intervals). Per-cell rows are descriptive; the weighted row is decisional (G_{all}, reading: inconclusive) and S1/S2 (both pass). Excl.: structurally excluded tasks (decision/trajectory count mismatch at search). Source: `analysis/fork/out/holdout_confirm_20261002T012353Z.md`.

K Withdrawn and Negative Results

W$\times$KV interaction (not supported). An earlier q4 analysis reported a W4$\times$KV4 interaction of -14.9 [-23.8, -6.0] pp ($k = 3$). Its KV16 reference lacked the Q/K rotation used for KV4, so rotation and quantization were confounded. With a matched rotated reference, the q4 interaction is -9.0 [-30.6, $+12.5$]. The pre-registered q8 replication gives $+2.1$ [-5.8, $+10.1$] over 6 blocks, crossed [-8.2, $+12.4$]. The replication is not supported, and we make no W$\times$KV claim.

Static precision restoration (appendix only). We tested whether restoring selected modules to higher precision recovers a damaged RTN-W3 model. We report this line as a mechanism stress test, not a method. In RTN-W3 the gains largely tracked the number of restored bits. A random MLP arm also recovered much of the gap, and no arm beat whole-model GPTQ-W4 at matched storage. The line was then paused and deferred. Its q8 RTN-W3 comparison of restored attention against byte-matched random attention was not completed: the attention arm and one random-attention seed ran, the remaining random-attention seeds are deferred, and no restoration analysis of these runs was checked, so we report no result from it.

Success equivalence. Not established (Table 3).

"Errors are absorbed". An interim reading of the injection study said that injected defects seldom change outcomes. The per-state analysis shows two-sided turnover (Appendix I), so this reading is retracted.

Superseded estimand names. The W89 draft reported $A = Y_{\mathrm{BB}} - Y_{\mathrm{QB}}$, $C = Y_{\mathrm{BB}} - Y_{\mathrm{BQ}}$ and I. In the present notation $A = \Delta_{\mathrm{B}} = F - H$ and $C = F - G$; the values are unchanged (Table 2).

Cell (weight)	states	G	H	F	$G - H$	mean I	off-diag. %	exp. %	κ	rev. % (n)
q4 GPTQ-W4 (1/2)	385	$-3.1\,[-9.0, +2.8]$	$+0.5\,[-5.4, +6.4]$	$+1.8\,[-4.8, +8.4]$	$-3.6\,[-9.1, +1.8]$	$-4.4\,[-12.5, +3.7]$	19.7	27.0	0.269	3.4 (13)
q8 GPTQ-W4 (1/4)	399	$-5.8\,[-10.2, -1.3]$	$-4.0\,[-10.2, +2.2]$	$-5.3\,[-11.6, +1.1]$	$-1.8\,[-6.4, +2.9]$	$-4.5\,[-10.4, +1.4]$	19.5	23.1	0.155	3.0 (12)
q8 AWQ-W4 (1/4)	396	$-6.6\,[-13.0, -0.1]$	$-8.3\,[-15.3, -1.3]$	$-8.3\,[-15.5, -1.2]$	$+1.8\,[-3.3, +6.8]$	$-6.6\,[-14.5, +1.4]$	23.7	26.0	0.088	1.0 (4)
Weighted pool	1180	$-4.6\,[-7.8, -1.5]$	$-2.8\,[-6.3, +0.6]$	$-2.5\,[-6.2, +1.3]$	$-1.8\,[-4.8, +1.1]$	$-5.0\,[-9.6, -0.4]$	20.7	25.7	0.195	2.7

Table 17: Telecom holdout, *post-hoc* R2 estimands on divergent states (descriptive; not part of the pre-registered confirmation). pp; crossed task×block 95% intervals; pool weights as pre-registered. Exp. and κ: marginal-preserving pairing agreement control; rev.: strict reversals. Source: `analysis/fork/out/holdout_r2_posthoc_20261002T024414Z.md`.